

Machine Learning — Repeat Exam

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Name:

Student ID:

Signature:

- Only write on the sheets given to you by supervisors. If you need more paper, ask the supervisors.
- Pages 16-18 can be used as scratch paper.
- All sheets (including scratch paper) have to be returned at the end.
- **Do not unstaple the sheets!**
- Wherever answer boxes are provided, please write your answers in them.
- Please write your student ID (*Matrikelnummer*) on every sheet you hand in.
- **Only use a black or a blue pen (no pencils, red or green pens!).**
- You are allowed to use your A4 sheet of handwritten notes (two sides). **No other materials (e.g. books, cell phones, calculators) are allowed!**
- Exam duration - 120 minutes.
- This exam consists of 21 pages, 11 problems. You can earn 54 points.

Probability distributions

For your reference, we provide the following probability distribution.

- Univariate normal distribution

$$\mathcal{N}(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- Bernoulli distribution

$$\text{Bern}(x|\theta) = \theta^x (1-\theta)^{(1-x)}$$

Student ID:

Decision Trees

Problem 1 [(2+2)=4 points] Assume you want to build a decision tree. Your data set consists of N samples, each with k features ($k \leq N$).

- a) If the features are binary, what is the maximum possible number of leaf nodes and the maximum depth of your decision tree?

Each feature can only be used once on a path from the root to the leaf.

Maximum number of leaf nodes: $\min(2^k, N)$ **1pt.**

Maximum depth: k **1pt.**

- b) If the features are continuous, what is the maximum possible number of leaf nodes and the maximum depth of your decision tree?

Each feature can be used multiple times on a path from the root to the leaf.

Maximum number of leaf nodes: N **1pt.**

Maximum depth: N **1pt.**

Regression

Problem 2 [(1+4)=5 points] We want to perform regression on a dataset consisting of N samples $\mathbf{x}_i \in \mathbb{R}^D$ with corresponding targets $y_i \in \mathbb{R}$ (represented compactly as $\mathbf{X} \in \mathbb{R}^{N \times D}$ and $\mathbf{y} \in \mathbb{R}^N$).

Assume that we have fitted an L_2 -regularized linear regression model and obtained the optimal weight vector $\mathbf{w}^* \in \mathbb{R}^D$ as

$$\mathbf{w}^* = \arg \min_{\mathbf{w}} \frac{1}{2} \sum_{i=1}^N (\mathbf{w}^T \mathbf{x}_i - y_i)^2 + \frac{\lambda}{2} \mathbf{w}^T \mathbf{w}$$

Note that there is no bias term.

Now, assume that we obtained a new data matrix $\mathbf{X}_{new}$ by scaling all samples by the same positive factor $a \in (0, \infty)$. That is, $\mathbf{X}_{new} = a\mathbf{X}$ (and respectively $\mathbf{x}_i^{new} = a\mathbf{x}_i$).

- a) Find the weight vector $\mathbf{w}_{new}$ that will produce the same predictions on $\mathbf{X}_{new}$ as $\mathbf{w}^*$ produces on $\mathbf{X}$.

Predictions of a linear regression model are generated as $\hat{y} = \mathbf{w}^T \mathbf{x}$.

This means that we need to ensure that $\mathbf{w}^{*T} \mathbf{x}_i = \mathbf{w}_{new}^T \mathbf{x}_i^{new}$ or equivalently $\mathbf{w}^{*T} \mathbf{x}_i = \mathbf{w}_{new}^T a \mathbf{x}_i$. Solving for $\mathbf{w}_{new}$ we get $\mathbf{w}_{new} = \frac{\mathbf{w}^*}{a}$

1 pt. for a correct answer.

- b) Find the regularization factor $\lambda_{new} \in \mathbb{R}$, such that the solution $\mathbf{w}_{new}^*$ of the new L_2 -regularized

linear regression problem

$$\mathbf{w}_{new}^* = \arg \min_{\mathbf{w}} \frac{1}{2} \sum_{i=1}^N (\mathbf{w}^T \mathbf{x}_i^{new} - y_i)^2 + \frac{\lambda_{new}}{2} \mathbf{w}^T \mathbf{w}$$

will produce the same predictions on $\mathbf{X}_{new}$ as $\mathbf{w}^*$ produces on $\mathbf{X}$.

Provide a mathematical justification for your answer.

The closed form solution for $\mathbf{w}^*$ on the original data $\mathbf{X}$ is

$$\mathbf{w}^* = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y}$$

0.5 pt.

The closed form solution for $\mathbf{w}_{new}^*$ on the new data $\mathbf{X}_{new}$ is

$$\begin{aligned} \mathbf{w}_{new}^* &= (\mathbf{X}_{new}^T \mathbf{X}_{new} + \lambda_{new} \mathbf{I})^{-1} \mathbf{X}_{new}^T \mathbf{y} \\ &= a(a^2 \mathbf{X}^T \mathbf{X} + \lambda_{new} \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y} \end{aligned}$$

1 pt.

by setting $\lambda_{new} = a^2 \lambda$, we get

$$\begin{aligned} &= a(a^2 \mathbf{X}^T \mathbf{X} + a^2 \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y} \\ &= \frac{1}{a} (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y} \\ &= \frac{1}{a} \mathbf{w}^* \end{aligned}$$

Which (according to our answer in part (a) of this problem) will produce the same predictions on $\mathbf{X}_{new}$ as $\mathbf{w}^*$ does on $\mathbf{X}$, as desired.

1 pt. for setting $\mathbf{w}_{new}^* = \frac{1}{a} \mathbf{w}^*$

1 pt. for correct (and justified) value of λ_{new}

Equivalent solution

$$\begin{aligned} \mathbf{w}_{new}^* &\stackrel{!}{=} \frac{\mathbf{w}^*}{a} = \frac{1}{a} \arg \min_{\mathbf{w}} \frac{1}{2} \sum_{i=1}^N (\mathbf{w}^T \mathbf{x}_i - y_i)^2 + \frac{\lambda}{2} \mathbf{w}^T \mathbf{w} \\ &= \frac{1}{a} \arg \min_{\mathbf{w}} \frac{1}{2} \sum_{i=1}^N \left(\frac{\mathbf{w}^T}{a} a \mathbf{x}_i - y_i \right)^2 + \frac{a^2 \lambda}{2} \frac{\mathbf{w}^T}{a} \frac{\mathbf{w}}{a} \\ &= \frac{a}{a} \arg \min_{\mathbf{w}_{new} = \frac{\mathbf{w}}{a}} \frac{1}{2} \sum_{i=1}^N (\mathbf{w}_{new}^T \mathbf{x}_i^{new} - y_i)^2 + \frac{a^2 \lambda}{2} \mathbf{w}_{new}^T \mathbf{w}_{new} \\ &\stackrel{!}{=} \mathbf{w}_{new}^* = \arg \min_{\mathbf{w}_{new}} \frac{1}{2} \sum_{i=1}^N (\mathbf{w}_{new}^T \mathbf{x}_i^{new} - y_i)^2 + \frac{\lambda_{new}}{2} \mathbf{w}_{new}^T \mathbf{w}_{new} \end{aligned}$$

1 pt. for setting $\mathbf{w}_{new}^* \stackrel{!}{=} \frac{\mathbf{w}^*}{a}$ and inserting a in the right spots (while keeping the loss function independent of a)

1 pt. for correctly substituting $\mathbf{w}_{new}$

For this equality to hold we need to match the regularization term by setting $\lambda_{new} = a^2\lambda$.

1 pt. for matching regularization term

1 pt. for correct (and justified) value of λ_{new}

Classification

Problem 3 [(1+2+3)=6 points] We would like to perform binary classification on multivariate binary data. That is, the data points $\mathbf{x}_i \in \{0, 1\}^D$ are binary vectors of length D , and each sample belongs to one of two classes $y_i \in \{1, 2\}$.

Consider the following generative classification model. We place a categorical prior on y

$$p(y = 1) = \pi_1 \quad p(y = 2) = \pi_2.$$

The class-conditional distributions are products of independent Bernoulli distributions

$$p(\mathbf{x} \mid y = 1, \boldsymbol{\alpha}) = \prod_{j=1}^D \text{Bern}(x_j \mid \alpha_j),$$

$$p(\mathbf{x} \mid y = 2, \boldsymbol{\beta}) = \prod_{j=1}^D \text{Bern}(x_j \mid \beta_j),$$

where $\boldsymbol{\alpha} \in [0, 1]^D$ and $\boldsymbol{\beta} \in [0, 1]^D$ are the respective parameter vectors for both classes. That is, each component x_j is distributed as $x_j \sim \text{Bern}(\alpha_j)$ if $y = 1$ or $x_j \sim \text{Bern}(\beta_j)$ if $y = 2$.

a) Write down the expression for the posterior distribution $p(y \mid \mathbf{x}, \boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\pi})$.

Using the Bayes formula

$$p(y \mid \mathbf{x}, \boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\pi}) = \frac{p(\mathbf{x} \mid y, \boldsymbol{\alpha}, \boldsymbol{\beta})p(y \mid \boldsymbol{\pi})}{p(\mathbf{x} \mid \boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\pi})}$$

$$= \frac{p(\mathbf{x} \mid y, \boldsymbol{\alpha}, \boldsymbol{\beta})p(y \mid \boldsymbol{\pi})}{p(\mathbf{x} \mid y = 1, \boldsymbol{\alpha}, \boldsymbol{\beta})p(y = 1 \mid \boldsymbol{\pi}) + p(\mathbf{x} \mid y = 2, \boldsymbol{\alpha}, \boldsymbol{\beta})p(y = 2 \mid \boldsymbol{\pi})}$$

b) Assume that $D = 3$, $\boldsymbol{\alpha} = [1/3, 1/3, 3/4]$, $\boldsymbol{\beta} = [2/3, 1/2, 1/2]$, $\pi_1 = 1/3$ and $\pi_2 = 2/3$.

Write down a data point $\mathbf{x}_1 \in \{0, 1\}^3$ that will be classified as class 1 by our model. Additionally, compute the posterior probability $p(y = 1 \mid \mathbf{x}_1, \boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\pi})$.

Consider $\mathbf{x}_1 = (0, 0, 1)^T$. The probability that $\mathbf{x}_1$ belongs to class 1 is

$$\begin{aligned}
 p(y = 1 \mid \mathbf{x}_1, \boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\pi}) &= \frac{p(\mathbf{x}_1 \mid y = 1, \boldsymbol{\alpha}, \boldsymbol{\beta})p(y = 1 \mid \boldsymbol{\pi})}{p(\mathbf{x}_1 \mid y = 1, \boldsymbol{\alpha}, \boldsymbol{\beta})p(y = 1 \mid \boldsymbol{\pi}) + p(\mathbf{x}_1 \mid y = 2, \boldsymbol{\alpha}, \boldsymbol{\pi})p(y = 2 \mid \boldsymbol{\pi})} \\
 &= \frac{p(\mathbf{x}_1 \mid y = 1, \boldsymbol{\alpha}, \boldsymbol{\beta})\pi_1}{p(\mathbf{x}_1 \mid y = 1, \boldsymbol{\alpha}, \boldsymbol{\beta})\pi_1 + p(\mathbf{x}_1 \mid y = 2, \boldsymbol{\alpha}, \boldsymbol{\pi})\pi_2} \\
 &= \frac{(1 - \alpha_1) \cdot (1 - \alpha_2) \cdot \alpha_3 \cdot \pi_1}{(1 - \alpha_1) \cdot (1 - \alpha_2) \cdot \alpha_3 \cdot \pi_1 + (1 - \beta_1) \cdot (1 - \beta_2) \cdot \beta_3 \cdot \pi_2} \\
 &= \frac{2/3 \cdot 2/3 \cdot 3/4 \cdot 1/3}{2/3 \cdot 2/3 \cdot 3/4 \cdot 1/3 + 1/3 \cdot 1/2 \cdot 1/2 \cdot 2/3} \\
 &= \frac{1/9}{1/9 + 1/18} \\
 &= \frac{2}{3}
 \end{aligned}$$

- c) Consider the case when $D = 2$, $\pi_1 = \pi_2 = 1/2$, and $\boldsymbol{\alpha} \in [0, 1]^2$ and $\boldsymbol{\beta} \in [0, 1]^2$ are known and fixed. Show that the resulting classification rule can be represented as a linear function of $\mathbf{x}$. That is, find $\mathbf{w} \in \mathbb{R}^2$ and $b \in \mathbb{R}$, such that

$$\{\mathbf{x} \in \{0, 1\}^2 : \mathbf{w}^T \mathbf{x} + b > 0\} = \{\mathbf{x} \in \{0, 1\}^2 : p(y = 1 \mid \mathbf{x}) > p(y = 2 \mid \mathbf{x})\}$$

$$\begin{aligned}
 &p(y = 1 \mid \mathbf{x}) > p(y = 2 \mid \mathbf{x}) \\
 \iff &p(\mathbf{x} \mid y = 1)p(y = 1) > p(\mathbf{x} \mid y = 2)p(y = 2) \\
 \iff &p(\mathbf{x} \mid y = 1) > p(\mathbf{x} \mid y = 2) \\
 \iff &\log p(\mathbf{x} \mid y = 1) > \log p(\mathbf{x} \mid y = 2) \\
 \iff &\log p(x_1 \mid y = 1) + \log p(x_2 \mid y = 1) > \log p(x_1 \mid y = 2) + \log p(x_2 \mid y = 2) \\
 \iff &x_1 \log \alpha_1 + (1 - x_1) \log(1 - \alpha_1) + x_2 \log \alpha_2 + (1 - x_2) \log(1 - \alpha_2) \\
 &\quad - x_1 \log \beta_1 - (1 - x_1) \log(1 - \beta_1) - x_2 \log \beta_2 - (1 - x_2) \log(1 - \beta_2) > 0 \\
 \iff &(\log \alpha_1 - \log(1 - \alpha_1) - \log \beta_1 + \log(1 - \beta_1))x_1 \\
 &\quad + (\log \alpha_2 - \log(1 - \alpha_2) - \log \beta_2 + \log(1 - \beta_2))x_2 \\
 &\quad + (\log(1 - \alpha_1) + \log(1 - \alpha_2) - \log(1 - \beta_1) - \log(1 - \beta_2)) > 0
 \end{aligned}$$

We can equivalently represent this inequality as $\mathbf{w}^T \mathbf{x} + b > 0$, where

$$\mathbf{w} = \begin{pmatrix} \log \alpha_1 - \log(1 - \alpha_1) - \log \beta_1 + \log(1 - \beta_1) \\ \log \alpha_2 - \log(1 - \alpha_2) - \log \beta_2 + \log(1 - \beta_2) \end{pmatrix} \in \mathbb{R}^2$$

and

$$b = (\log(1 - \alpha_1) + \log(1 - \alpha_2) - \log(1 - \beta_1) - \log(1 - \beta_2)) \in \mathbb{R}$$

Kernels

Problem 4 [(4)=4 points] Prove or disprove whether the following operations on sets $A, B \subseteq \mathcal{X}$, where $\mathcal{X}$ is a finite set, define a valid kernel.

- a) $k(A, B) = |A \times B|$, where $A \times B = \{(a, b) : a \in A, b \in B\}$ denotes the cartesian product and $|S|$ denotes the cardinality of set S , i.e. the number of elements in S .

Yes, using cardinality as a feature map $|A \times B| = |A| \cdot |B|$ obviously defines a kernel.

0.5pt. for definition of a kernel or stating Mercer's theorem (only once and only if no subtask was solved).

1pt. for a correct answer.

b) $k(A, B) = |A \cap B|$

Since the set $\mathcal{X}$ is finite we can denote its members as x_i , with $i \in \{1, 2, \dots, |\mathcal{X}|\}$, and define the membership vector $\mathbf{a}$, with

$$a_i = \begin{cases} 1 & \text{if } x_i \in A, \\ 0 & \text{otherwise.} \end{cases}$$

Using this as a feature map we can write

$$|A \cap B| = \mathbf{a}^T \mathbf{b},$$

showing that $|A \cap B|$ is a kernel.

0.5pt. for definition of a kernel or stating Mercer's theorem (only once and only if no subtask was solved).

1pt. for a correct answer.

c) $k(A, B) = |A \cup B|$

Looking at the determinant of a Gram matrix defined by two objects $A, B \subseteq \mathcal{X}$ we have

$$\det \begin{pmatrix} |A| & |A \cup B| \\ |A \cup B| & |B| \end{pmatrix} = |A||B| - |A \cup B|^2,$$

which is negative if $A \neq B$. Hence, the Gram matrix is not positive semidefinite and according to Mercer's theorem $|A \cup B|$ is not a kernel.

0.5pt. for definition of a kernel or stating Mercer's theorem (only once and only if no subtask was solved).

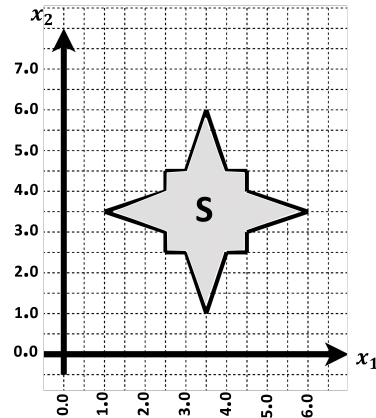
2pt. for a correct counterexample.

Optimization

Problem 5 [(1+3+2)=6 points] Let f be the following convex function on $\mathbb{R}^2$:

$$f(x_1, x_2) = e^{x_1+x_2} - 5 \cdot \log(x_2)$$

a) Consider the following shaded region $\mathbf{S}$ in $\mathbb{R}^2$. Is this region convex? Why?

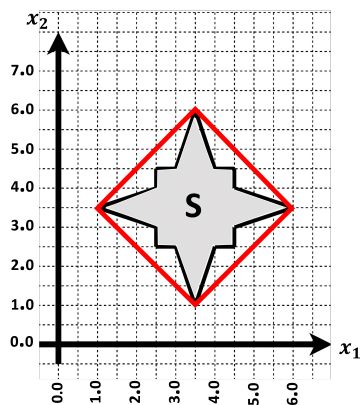


It is not. Because we can choose two points in S where the line connecting the points does not completely reside in S .

1pt. for a correct answer and explanation.

b) Find the maximizer (x_1^*, x_2^*) of f over the shaded region S . For your computations, you can pick values from the following table. Justify your answer.

$e^{4.5} = 90.017$	$e^{5.0} = 148.41$	$e^{5.5} = 244.69$	$e^{6.5} = 665.14$
$e^{7.0} = 1096.63$	$e^{7.5} = 1808.04$	$e^{8.0} = 2980.95$	$e^{8.5} = 4914.76$
$e^{9.0} = 8103.08$	$e^{9.5} = 13359.726$	$e^{10.0} = 22026.46$	$e^{10.5} = 36315.50$
$\log(1.0) = 0$	$\log(2.5) = 0.9162$	$\log(3.0) = 1.0986$	$\log(3.5) = 1.2527$
$\log(4.0) = 1.3862$	$\log(4.5) = 1.5040$	$\log(5.0) = 1.6094$	$\log(6.0) = 1.7917$



According to the lecture, we need to compute the value of f only on the four corners of the convex hull. Afterwards, we pick the corner at which the value of f is larger.

$$\begin{aligned}
 f(1, 3.5) &= e^{4.5} - 5 \cdot \log(3.5) = 83.7533 & f(3.5, 1.0) &= e^{4.5} - 5 \cdot \log(1.0) = 90.01 \\
 f(6, 3.5) &= e^{9.5} - 5 \cdot \log(3.5) = 13353.46 & f(3.5, 6.0) &= e^{9.5} - 5 \cdot \log(6.0) = 13350.76
 \end{aligned}$$

Therefore, the maximum of f occurs at point $(6, 3.5)$, and the maximum of f is 13353.46.

1pt. for writing down the correct answer.

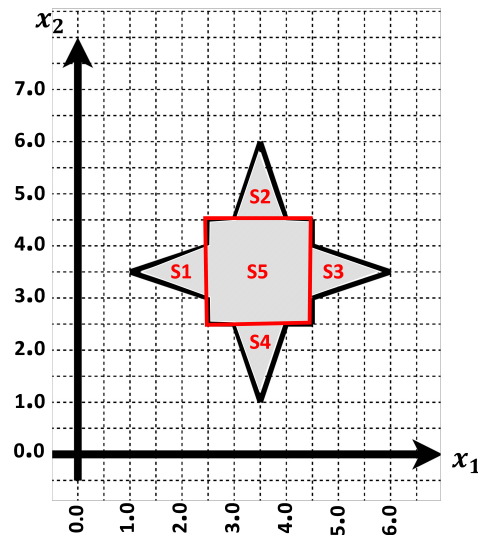
2pt. for completely correct explanation (checking only the 4 corners of the convex-hull).

1pt. for partially correct explanation (checking all 16 corners, gradient ascent, etc.)

- c) Assume that we are given an algorithm $\text{ConvOpt}(f, \mathcal{X})$ that takes as input a convex function f and any convex region $\mathcal{X}$, and returns the minimum of f over $\mathcal{X}$.

Using the ConvOpt algorithm, how would you find the global minimum of f over the shaded region S ?

We can partition the shaded region S to the following five convex regions. Afterwards, we run the ConvOpt algorithm separately for the 5 regions.



- 2pt.** for a feasible approach.
1pt. for an approach with a few addressable issues.

SVM

Problem 6 [(5)=5 points] Given the data points

$$\mathbf{x}_1 = (1, 1, 0, 1)^T \quad \mathbf{x}_2 = (1, 1, 1, 0)^T \quad \mathbf{x}_3 = (0, 1, 1, 1)^T \quad \mathbf{x}_4 = (0, 0, 1, 1)^T$$

Prove or disprove whether the following combinations of labels $\mathbf{y}$ and dual variables $\boldsymbol{\alpha}$ are the optimal solutions of a soft-margin SVM with $C = 1$.

- a) $\mathbf{y} = (-1, -1, 1, 1)^T$, $\boldsymbol{\alpha} = (0.6, 0.6, 1, 0)^T$
- b) $\mathbf{y} = (-1, -1, 1, 1)^T$, $\boldsymbol{\alpha} = (\frac{2}{3}, \frac{2}{3}, \frac{4}{3}, 0)^T$
- c) $\mathbf{y} = (-1, 1, -1, 1)^T$, $\boldsymbol{\alpha} = (1, 1, 1, 1)^T$

a) No, $\sum_i \alpha_i y_i \neq 0$. **1pt.** (**0.5pt.** for constraint)

b) No, $\alpha_3 > C$. **1pt.** (**0.5pt.** for constraint)

c) This solution is feasible, so we need to check optimality. The dual problem of the soft-margin SVM is

$$g(\boldsymbol{\alpha}) = \sum_i \alpha_i - \frac{1}{2} \sum_i \sum_j y_i y_j \alpha_i \alpha_j \mathbf{x}_i^T \mathbf{x}_j.$$

0.5pt., also for a correct approach for deriving it.

To check whether this is an optimal solution we first look at the gradient, which is given by

$$(\nabla_{\boldsymbol{\alpha}} g(\boldsymbol{\alpha}))_i = 1 - \sum_j y_i y_j \alpha_j \mathbf{x}_i^T \mathbf{x}_j.$$

1pt.

To calculate the gradient, we need the kernel matrix, with $K_{ij} = \mathbf{x}_i^T \mathbf{x}_j$:

$$K = \begin{pmatrix} 3 & 2 & 2 & 1 \\ 2 & 3 & 2 & 1 \\ 2 & 2 & 3 & 2 \\ 1 & 1 & 2 & 2 \end{pmatrix}.$$

0.5pt.

Using this we can directly calculate

$$\partial_{\alpha_1} g(\boldsymbol{\alpha}) = 1 - (3 - 2 + 2 - 1) = -1$$

$$\partial_{\alpha_2} g(\boldsymbol{\alpha}) = 1 - (-2 + 3 - 2 + 1) = 1$$

$$\partial_{\alpha_3} g(\boldsymbol{\alpha}) = 1 - (2 - 2 + 3 - 2) = 0$$

$$\partial_{\alpha_4} g(\boldsymbol{\alpha}) = 1 - (-1 + 1 - 2 + 2) = 1$$

0.5pt.

Now let us consider the constraints. Since all $\alpha_i = 1 = C$ we cannot increase any α_i . However, we also can't decrease α_1 without decreasing either α_2 or α_4 due to the constraint $\sum_i \alpha_i y_i = 0$. The absolute values of these three derivatives are equally high, which means that decreasing these values would not increase $g(\boldsymbol{\alpha})$. Hence, we cannot increase $g(\boldsymbol{\alpha})$ any further and this is indeed an optimal solution. **0.5pt.**

Deep Learning

Problem 7 [(2+2)=4 points] You are trying to solve a regression task and you want to choose between two approaches:

1. A simple linear regression model.
2. A feed forward neural network $f_{\mathbf{W}}(\mathbf{x})$ with L hidden layers, where each hidden layer $l \in \{1, \dots, L\}$ has a weight matrix $\mathbf{W}_l \in \mathbb{R}^{D \times D}$ and a ReLU activation function. The output layer has a weight matrix $\mathbf{W}_{L+1} \in \mathbb{R}^{D \times 1}$ and no activation function.

In both models, there are no bias terms.

Your dataset $\mathcal{D}$ contains data points with nonnegative features $\mathbf{x}_i$ and the target y_i is continuous:

$$\mathcal{D} = \{\mathbf{x}_i, y_i\}_{i=1}^N, \quad \mathbf{x}_i \in \mathbb{R}_{\geq 0}^D, \quad y_i \in \mathbb{R}$$

Let $\mathbf{w}_{LS}^* \in \mathbb{R}^D$ be the optimal weights for the linear regression model corresponding to a global minimum of the following least squares optimization problem:

$$\mathbf{w}_{LS}^* = \arg \min_{\mathbf{w} \in \mathbb{R}^D} \mathcal{L}_{LS}(\mathbf{w}) = \arg \min_{\mathbf{w} \in \mathbb{R}^D} \frac{1}{2} \sum_{i=1}^N (\mathbf{w}^T \mathbf{x}_i - y_i)^2$$

Let $\mathbf{W}_{NN}^* = \{\mathbf{W}_1^*, \dots, \mathbf{W}_{L+1}^*\}$ be the optimal weights for the neural network corresponding to a global minimum of the following optimization problem:

$$\mathbf{W}_{NN}^* = \arg \min_{\mathbf{W}} \mathcal{L}_{NN}(\mathbf{W}) = \arg \min_{\mathbf{W}} \frac{1}{2} \sum_{i=1}^N (f_{\mathbf{W}}(\mathbf{x}_i) - y_i)^2$$

- a) Assume that the optimal $\mathbf{W}_{NN}^*$ you obtain are non-negative.
 What will be the relation ($<$, $\leq$, $=$, $\geq$, $>$) between the neural network loss $\mathcal{L}_{NN}(\mathbf{W}_{NN}^*)$ and the linear regression loss $\mathcal{L}_{LS}(\mathbf{w}_{LS}^*)$? Provide a mathematical argument to justify your answer.

Note that for any non-negative $\mathbf{x}$ and any non-negative $\mathbf{W}$ it holds $\text{ReLU}(\mathbf{x}\mathbf{W}) = \mathbf{x}\mathbf{W}$.

1pt: Therefore, since our data points have non-negative features $\mathbf{x}_i$ and the optimal weights $\mathbf{W}_{NN}^*$ are non-negative, every ReLU layer is equivalent to a linear layer when plugging in the optimal weights. This means we can write

$$\begin{aligned} f_{\mathbf{W}_{NN}^*}(\mathbf{x}_i) &= \text{ReLU}(\text{ReLU}(\text{ReLU}(\mathbf{x}_i^T \mathbf{W}_1^*) \mathbf{W}_2^*) \cdots \mathbf{W}_L^*) \mathbf{W}_{L+1}^* \\ &= \mathbf{x}_i^T \mathbf{W}_1^* \mathbf{W}_2^* \cdots \mathbf{W}_{L+1}^* \\ &= \mathbf{x}_i^T \mathbf{w}_{NN}^* \end{aligned}$$

where we defined $\mathbf{w}_{NN}^* = \mathbf{W}_1^* \mathbf{W}_2^* \cdots \mathbf{W}_{L+1}^*$. From this we can see that the neural network with optimal weights behaves like a linear regression with a different set of weights $\mathbf{w}_{NN}^*$.

-0.5pt: if mentioned only non-negative weights or features but not both

-0.5pt: if the answer under (a) is actually the correct answer for (b) and vice versa

1pt: Note also that linear regression is a special case of the above neural network, i.e. for any weights $\mathbf{w}_{LS}$ you can find weights $\mathbf{W}_{NN}$ that produce the same output.

Given the above facts and since we the optimal weights correspond to a global minima we can conclude that $\mathcal{L}_{NN}(\mathbf{W}_{NN}^*) = \mathcal{L}_{LS}(\mathbf{w}_{LS}^*)$ and the optimal weights found by solving the least squares optimization problem will be $\mathbf{w}_{LS}^* = \mathbf{w}_{NN}^*$.

- b) In contrast to (a), now assume that the optimal weights $\mathbf{w}_{LS}^*$ you obtain are non-negative. What will be the relation ($<, \leq, =, \geq, >$) between the linear regression loss $\mathcal{L}_{LS}(\mathbf{w}_{LS}^*)$ and the neural network loss $\mathcal{L}_{NN}(\mathbf{W}_{NN}^*)$? Provide a mathematical argument to justify your answer.

1pt: As stated in (a) linear regression is a special case of the above neural network, i.e. for any weights $\mathbf{w}_{LS}$ you can find weights $\mathbf{W}_{NN}$ that produce the same output. That is, everything that can be learned with a linear regression can be learned equally well with a neural network.

1pt: However, the reverse direction doesn't hold, since in principle neural networks can learn more complicated functions compared to linear regression. Moreover, the given fact that $\mathbf{w}_{LS}^*$ are non-negative does not tell us anything about the optimal weights of the neural network $\mathbf{W}_{NN}^*$.

Therefore it holds $\mathcal{L}_{NN}(\mathbf{W}_{NN}^*) \leq \mathcal{L}_{LS}(\mathbf{w}_{LS}^*)$ since the neural network can potentially find a better fit for the data (e.g. by taking advantage of non-linearity).

-0.25pt: for stating $<$ instead of $\leq$

Dimensionality Reduction

Problem 8 [(3+2+2)=7 points] You are given $N = 4$ data points: $\{\mathbf{x}_i\}_{i=1}^4, \mathbf{x}_i \in \mathbb{R}^3$, represented with the matrix $\mathbf{X} \in \mathbb{R}^{4 \times 3}$.

$$\mathbf{X} = \begin{bmatrix} 4 & 3 & 2 \\ 2 & 1 & -2 \\ 4 & -1 & 2 \\ -2 & 1 & 2 \end{bmatrix}$$

Hint: In this task the results of all (final and intermediate) computations happen to be integers.

- a) Perform principal component analysis (PCA) of the data $\mathbf{X}$, i.e. find the principal components and their associated variances in the transformed coordinate system. Show your work.

First we center the data. The mean is $\bar{\mathbf{x}} = [2, 1, 1]$, thus we have

$$\mathbf{X}_c = \mathbf{X} - \bar{\mathbf{x}} = \begin{bmatrix} 2 & 2 & 1 \\ 0 & 0 & -3 \\ 2 & -2 & 1 \\ -4 & 0 & 1 \end{bmatrix}$$

Then we compute the covariance matrix.

$$\Sigma_{X_c} = \frac{1}{N} \mathbf{X}_c^T \mathbf{X}_c = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Since Σ_{X_c} it is already in a diagonal form we can conclude that $\mathbf{\Lambda} = \Sigma_{X_c}$ and $\mathbf{\Gamma} = \mathbf{I}_3$, and that it holds $\Sigma_{X_c} = \mathbf{\Gamma} \mathbf{\Lambda} \mathbf{\Gamma}^T$. The principal components are the canonical basis vectors.

- 1 pt.** for centring the data correctly or computing the correct mean
- 1 pt.** for computing correctly the covariance matrix.
- 1 pt.** for giving the correct transformed coordinate system.
- 0.5 pt** if no centring.

- b) Project the data to two dimensions, i.e. write down the transformed data matrix $\mathbf{Y} \in \mathbb{R}^{4 \times 2}$ using the top-2 principal components you computed in (a). What fraction of variance of $\mathbf{X}$ is preserved by $\mathbf{Y}$?

The projection matrix is:

$$\mathbf{\Gamma}_{trunc} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}$$

since we pick the first and the third principal vector corresponding to the two largest eigenvalues. Thus, we have

$$\mathbf{Y} = \mathbf{X}\mathbf{\Gamma}_{trunc} = \begin{bmatrix} 2 & 1 \\ 0 & -3 \\ 2 & 1 \\ -4 & 1 \end{bmatrix}$$

We preserve $\frac{6+3}{6+2+3} = \frac{9}{11}$ of the variance.

- 0.5 pt.** for correct computation for projection of $\mathbf{X}$ or $\mathbf{X}_c$ based on the results obtained in question a)
- 0.5 pt.** for correct computation for the fraction of variance based on the results obtained in question a)
- 0.5 pt.** for correct result for projection of $\mathbf{X}$ or $\mathbf{X}_c$
- 0.5 pt.** for correct result for the fraction of variance

- c) Let $\mathbf{x}_5 \in \mathbb{R}^3$ be a new data point. Specify the vector $\mathbf{x}_5$ such that performing PCA on the data including the new data point $\{\mathbf{x}_i\}_{i=1}^5$ leads to exactly the same principal components as in (a).

Let $\mathbf{x}_5 = \bar{\mathbf{x}}$, i.e. the new data point equals the mean before including $\mathbf{x}_5$ to the dataset. Therefore, the new mean including $\mathbf{x}_5$ is equal to the old mean. We have:

$$\mathbf{X}_c = \mathbf{X} - \bar{\mathbf{x}} = \begin{bmatrix} 2 & 2 & 1 \\ 0 & 0 & -3 \\ 2 & -2 & 1 \\ -4 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

which leads to the same $\Sigma_{\mathbf{X}_c}$ as in (a) up to a difference in the multiplicative constant, in (a) we had $\frac{1}{4}\mathbf{X}_c^T\mathbf{X}_c$ and here we have $\frac{1}{5}\mathbf{X}_c^T\mathbf{X}_c$. While this difference leads to different eigenvalues, the eigenvectors and thus the principal components stay the same.

- 2 pt.** for saying $\mathbf{x}_5 = \bar{\mathbf{x}}$
- 1 pt.** for only reasoning on the shifted data

Clustering

Problem 9 [(4)=4 points] Let $\mu_1, \dots, \mu_K$ be the centroids computed by the K -means algorithm. Prove that the set $\mathcal{X}_j$ of all points in $\mathbb{R}^D$ assigned during inference to the cluster j is a convex set.

$$\mathcal{X}_j := \{\mathbf{x} \in \mathbb{R}^D : \mathbf{x} \text{ would be assigned to centroid } \mu_j \text{ by } K\text{-means}\}$$

Hint: start by thinking about the case with $K = 2$.

Bonus (applicable for arbitrary solutions):

1pt: in case of only two clusters $\mathcal{X}_j$ is a **half-space (0.5pts)** since the decision boundary is a hyperplane and therefore **convex (0.5pts)** if followed from a valid statement, otherwise **0pts)**

1.5pts for the formal definition of $\mathcal{X}_j$ as an intersection of $K - 1$ sets

$$\begin{aligned}\mathcal{X}_j &= \bigcap_{k \neq j} \{\mathbf{x} \in \mathbb{R}^D \mid \|\mathbf{x} - \mu_j\|^2 < \|\mathbf{x} - \mu_k\|^2\} \\ &= \bigcap_{k \neq j} \{\mathbf{x} \in \mathbb{R}^D \mid (\mu_k - \mu_j)^T (\mathbf{x} - (\mu_k + \mu_j)/2) < 0\}\end{aligned}$$

0.5pts for the definition of convexity (if applied during further argumentation, otherwise **0pts)**

0.5pts for a graphical representation of the general setting for K -means (if used for further argumentation, otherwise **0pts)**

Solution 1:

In general for K clusters $\mathcal{X}_j$ is an **intersection of $K - 1$ half-spaces (3pts)**:

- this point has to be clarified to get 3pts, otherwise only **1.5pts for stating this fact**,
- any complete and understandable **justification counts for 1.5pts**, for example
- consider the case $K = 2$ where $\mathcal{X}_j$ is a half-space and generalize it for larger K or
- rewrite the definition of $\mathcal{X}_j$ as above in form of an intersection of $K - 1$ half-spaces.

Then $\mathcal{X}_j$ is convex since arbitrary **intersections of convex sets** are convex (**1pts** for any correct justification of convexity here).

Solution 2 (directly apply the definition of convexity):

Given $\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{X}_j$ and $\lambda \in (0, 1)$ we show that $\mathbf{x}_\lambda = \lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2 \in \mathcal{X}_j$ (**0.5pts** for the definition of convexity).

We **rewrite the definition of $\mathcal{X}_j$ (1.5pts)** as

$$\mathcal{X}_j = \{\mathbf{x} \in \mathbb{R}^D \mid \|\mathbf{x} - \mu_j\|^2 < \|\mathbf{x} - \mu_k\|^2 \text{ for all } k \neq j\}$$

2pts for the proof of “ $\mathbf{x}_\lambda \in \mathcal{X}_j$ ” itself if completely correct (**1pt** if contains minor mistakes). For example: note, that for each $k \neq j$ the following conditions are equivalent (since $\|a\|^2 - \|b\|^2 = (a - b)^T(a + b)$)

$$\|\mathbf{x} - \mu_j\|^2 < \|\mathbf{x} - \mu_k\|^2 \Leftrightarrow (\mu_k - \mu_j)^T (\mathbf{x} - (\mu_k + \mu_j)/2) < 0$$

and are satisfied for $\mathbf{x}_1$ and $\mathbf{x}_2$. Then we can conclude for $\mathbf{x}_\lambda$ that

$$\begin{aligned}
 (\boldsymbol{\mu}_k - \boldsymbol{\mu}_j)^T (\mathbf{x}_\lambda - (\boldsymbol{\mu}_k + \boldsymbol{\mu}_j)/2) &= (\boldsymbol{\mu}_k - \boldsymbol{\mu}_j)^T (\lambda \mathbf{x}_1 - \lambda(\boldsymbol{\mu}_k + \boldsymbol{\mu}_j)/2 + (1 - \lambda)\mathbf{x}_2 - (1 - \lambda)(\boldsymbol{\mu}_k + \boldsymbol{\mu}_j)/2) \\
 &= \lambda \underbrace{(\boldsymbol{\mu}_k - \boldsymbol{\mu}_j)^T (\mathbf{x}_1 - (\boldsymbol{\mu}_k + \boldsymbol{\mu}_j)/2)}_{<0} + (1 - \lambda) \underbrace{(\boldsymbol{\mu}_k - \boldsymbol{\mu}_j)^T (\mathbf{x}_2 - (\boldsymbol{\mu}_k + \boldsymbol{\mu}_j)/2)}_{<0} \\
 &< 0
 \end{aligned}$$

Problem 10 [(2)=2 points] Given three 1-dimensional Gaussian distributions $\mathcal{N}(\mu_i, \sigma_i^2)$ with parameters

$$\begin{array}{lll} \mu_1 = 1, & \mu_2 = -1, & \mu_3 = 0, \\ \sigma_1 = 1, & \sigma_2 = 0.5, & \sigma_3 = 2.5 \end{array}$$

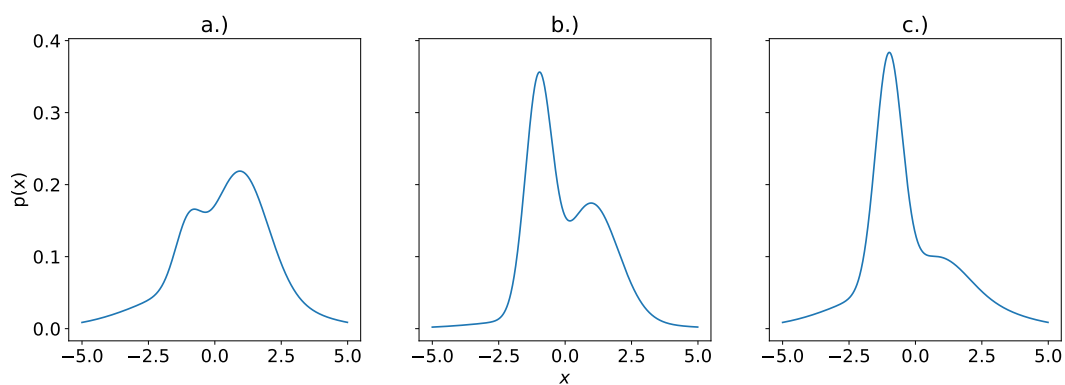
and three different vectors of mixing coefficients π defining categorical cluster priors.

Match the value of π in each row of the following table with one of the probability density functions

$$p(x) = \sum_{i=1}^3 \pi_i \mathcal{N}(x | \mu_i, \Sigma_i)$$

of the resulting GMM showed below. Fill in the last column of the table, no argumentation required.

	π_1	π_2	π_3	PDF (a, b or c)
case 1	0.111...	0.444...	0.444...	
case 2	0.444...	0.111...	0.444...	
case 3	0.444...	0.444...	0.111...	



case 1 is c.)

case 2 is a.)

case 3 is b.)

1 pt. if at least one case is correct.

2 pts. if all cases are correctly matched.

Variational Inference

Problem 11 [(3+1+1+2)=7 points] Consider the following latent variable probabilistic model

$$\begin{aligned} p(z) &= \mathcal{N}(z \mid 0, 1) \\ p(x \mid z) &= \mathcal{N}(x \mid z, 1) \end{aligned}$$

We want to approximate the posterior distribution $p(z \mid x)$ using the following variational family

$$\mathcal{Q} = \{\mathcal{N}(z \mid \mu, 1) \text{ for } \mu \in \mathbb{R}\}$$

that includes all normal distributions with unit variance.

Questions (a), (b), (c) and (d) are all concerning this setup.

Hint: Variance of $p(z \mid x)$ is equal to 0.5.

- a) Write down the closed-form expression for ELBO $\mathcal{L}(q)$ and simplify it. You can ignore all the terms constant in μ .

$$\begin{aligned} \mathcal{L}(q) &= \mathbb{E}_q [\log p(x, z) - \log q(z \mid \mu)] && \mathbf{0.5pt.} \text{ for definition of ELBO} \\ &= \mathbb{E}_q [\log p(x \mid z) + \log p(z) - \log q(z \mid \mu)] && \mathbf{0.5pt.} \text{ for expanding } \log p(x, z) \\ &= \mathbb{E}_q \left[-\frac{1}{2}(x - z)^2 - \frac{1}{2}z^2 + \frac{1}{2}(z - \mu)^2 \right] + \text{const.} && \mathbf{1pt.} \text{ for plugging in densities} \\ &= \mathbb{E}_q \left[zx - \frac{1}{2}z^2 - z\mu + \frac{1}{2}\mu^2 \right] + \text{const.} \\ &= \mathbb{E}_q [z] x - \frac{1}{2}\mathbb{E}_q [z^2] - \mathbb{E}_q [z] \mu + \frac{1}{2}\mu^2 + \text{const.} && \mathbf{0.5pt.} \text{ for linearity of expectation} \\ &= \mu x - \frac{1}{2}(\mu^2 + 1^2) - \mu^2 + \frac{1}{2}\mu^2 + \text{const.} \\ &= \mu x - \frac{1}{2}(\mu^2 + 1^2) - \mu^2 + \frac{1}{2}\mu^2 + \text{const.} \\ &= \mu x - \mu^2 + \text{const.} && \mathbf{0.5pt.} \text{ for simplifying} \end{aligned}$$

- b) Find the optimal variational distribution $q^* \in \mathcal{Q}$ that maximizes the ELBO

$$q^* = \arg \max_{q \in \mathcal{Q}} \mathcal{L}(q)$$

i.e. find the mean μ^* of the optimal variational distribution q^* .

We already know the ELBO: just compute the derivative w.r.t. μ and set it to zero to obtain the optimal μ^* .

$$\begin{aligned} \mathcal{L}(\mu) &= \mu x - \mu^2 + \text{const.} \\ \frac{\partial}{\partial \mu} \mathcal{L}(\mu) &= x - 2\mu \stackrel{!}{=} 0 \\ \iff \mu^* &= \frac{x}{2} \end{aligned}$$

1 pt. for setting the gradient of ELBO to zero and solving for μ .

c) Assume that the optimal q^* (i.e., the optimal μ^*) from question (b) is given. Which of the following statements is true?

(1) $\mathbb{KL}(q(z | \mu^*) \parallel p(z | x)) < 0$

(2) $\mathbb{KL}(q(z | \mu^*) \parallel p(z | x)) = 0$

(3) $\mathbb{KL}(q(z | \mu^*) \parallel p(z | x)) > 0$

Justify your answer.

KL-divergence is non-negative, so (1) is impossible. KL-divergence is equal to zero if and only if two distributions are equal. We know that the variance of $p(z | x)$ is 0.5 (see hint), and the variance of $q(z | \mu)$ for any μ is equal to 1 (by definition). Hence, $q(z | \mu^*) \neq p(z | x)$ and therefore KL-divergence is greater than zero (option (3)).

1 pt. for the correct answer with justification.

d) For each of the conditions (1), (2), (3) from question (c) above, provide a parametric variational family $\mathcal{Q}_i$, such that the optimal q_i^* from each family would fulfill the respective condition, or explain why it's impossible.

That is, provide $\mathcal{Q}_1$, such that for $q_1^* = \arg \max_{q \in \mathcal{Q}_1} \mathcal{L}(q)$ we have $\mathbb{KL}(q_1^*(z) \parallel p(z | x)) < 0$, for $q_2^* = \arg \max_{q \in \mathcal{Q}_2} \mathcal{L}(q)$ we have $\mathbb{KL}(q_2^*(z) \parallel p(z | x)) = 0$, and for $q_3^* = \arg \max_{q \in \mathcal{Q}_3} \mathcal{L}(q)$ we have $\mathbb{KL}(q_3^*(z) \parallel p(z | x)) > 0$.

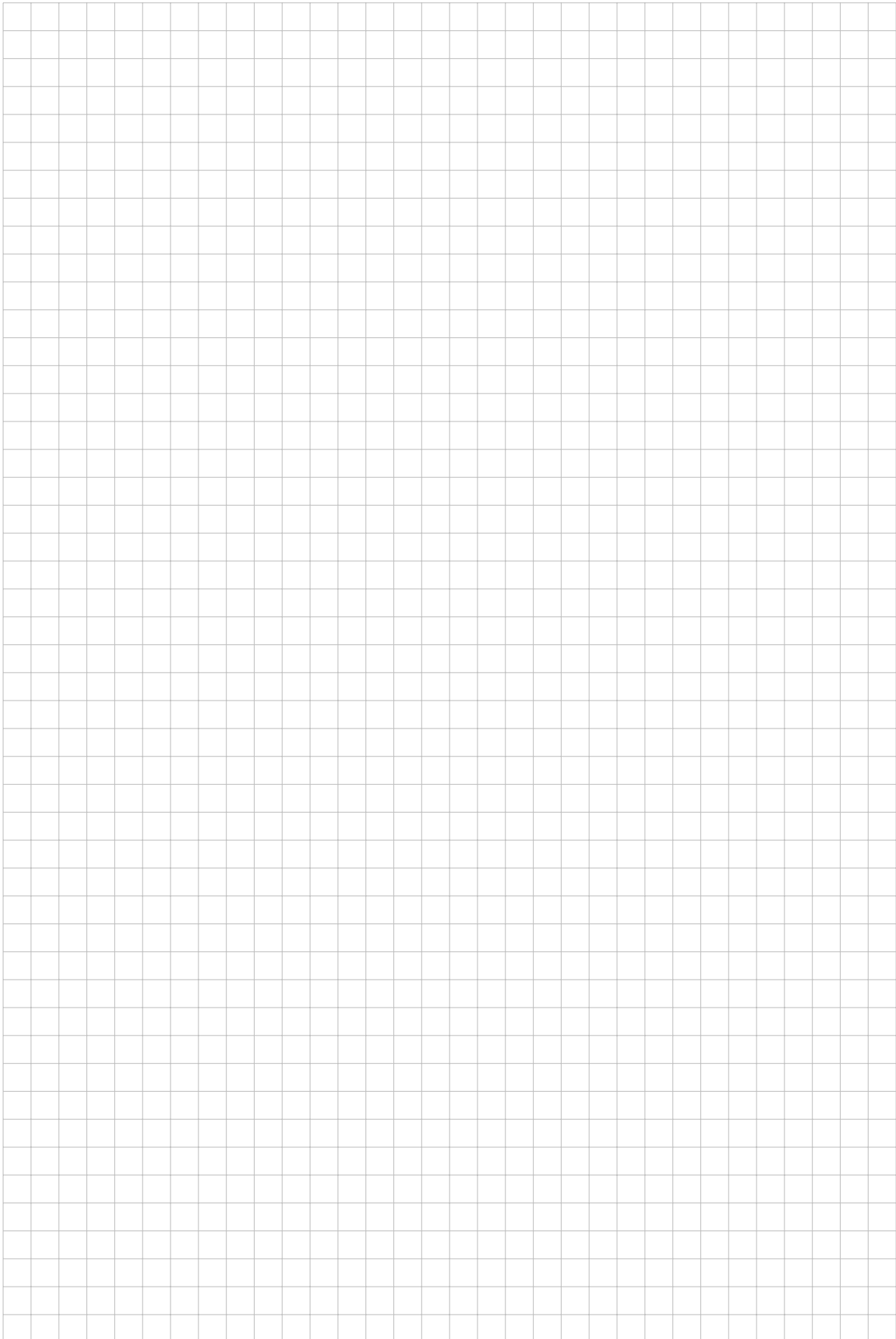
(1) Is impossible since KL is always nonnegative. **0.5 pt.**

(2) The likelihood $p(x | z)$ and the prior $p(z)$ are both normal distributions, hence the posterior $p(z | x)$ is also a normal distribution. Therefore, the posterior is contained in the set of all normal distributions $\mathcal{Q}_2 = \{\mathcal{N}(\mu, \sigma^2) \text{ for } \mu \in \mathbb{R}, \sigma^2 \in \mathbb{R}_{>0}\}$. **1 pt.**

(3) $\mathcal{Q}_3 = \{\mathcal{N}(\mu, 1) \text{ for } \mu \in \mathbb{R}\}$ works (see answer to (c)). **0.5 pt.**



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